Module I: Android Fundamentals & UI Design

Android Basics and Resources

1. Develop a simple "Hello World" Android application demonstrating project creation, Android Studio interface, Android components, AVD creation, and application execution.
2. Develop an Android application demonstrating the use of Programming Resources (colors.xml, themes.xml, strings.xml, dimens.xml, drawable resources) and apply them in UI components.
3. Develop an Android application demonstrating complete Activity Lifecycle methods and analyze execution using Logcat.

Fragments and Activity Interaction

4. Develop an Android application to create and integrate a Fragment and demonstrate Fragment lifecycle methods.
5. Develop an Android application implementing Fragment-based UI navigation between multiple screens.
6. Develop an Android application demonstrating communication between Activity and Fragment.

Basic Layout Management

7. Develop an Android application using LinearLayout (Horizontal and Vertical) to design structured UI.
8. Develop an Android application using RelativeLayout to align UI components dynamically.
9. Develop an Android application using TableLayout to design a structured Login/Registration Form.

Advanced Layout and View Handling

10. Develop an Android application using ScrollView to display long scrollable content.
11. Develop an Android application using AbsoluteLayout demonstrating fixed positioning.
12. Develop an Android application using FrameLayout to overlay UI elements.

Adapter Based Views

13. Develop an Android application using ListView with ArrayAdapter and implement item click handling.
14. Develop an Android application using GridView with Custom Adapter
15. Develop an Android application implementing RecyclerView as a modern alternative to ListView.

Module 2: Android System Components & Data Handling

Menus, Dialogs and UI Interaction

1. Develop an Android application implementing AppBar (Toolbar) and Options Menu.
2. Develop an Android application demonstrating AlertDialog and custom DialogFragment.
3. Develop an Android application implementing contextual menus and popup menus.

Intents, Events and Listeners

4. Develop an Android application implementing Explicit and Implicit Intents.
5. Develop an Android application demonstrating Event Handling using Listeners (OnClick, OnLongClick, etc.).
6. Develop an Android application using Adapters to populate Spinner or ListView dynamically.

Services, Notifications and Broadcast Receivers

7. Develop an Android application implementing Started and Bound Services.
8. Develop an Android application implementing Notifications using Notification Manager.
9. Develop an Android application implementing Static and Dynamic Broadcast Receivers.

Database and Networking

10. Develop an Android application implementing SQLite database CRUD operations.
11. Develop an Android application integrating SQLite with ListView or RecyclerView.
12. Develop an Android application to fetch and parse JSON data from a web

Concurrency, Media and Security

13. Develop an Android application demonstrating Threads and Handlers for background processing.
14. Develop an Android application implementing Media APIs and Telephone APIs.
15. Develop an Android application implementing Runtime Permissions and security handling in AndroidManifest.xml.

Practical 01

Develop a simple "Hello World" Android application demonstrating project creation, Android Studio interface, Android components, AVD creation, and application execution.

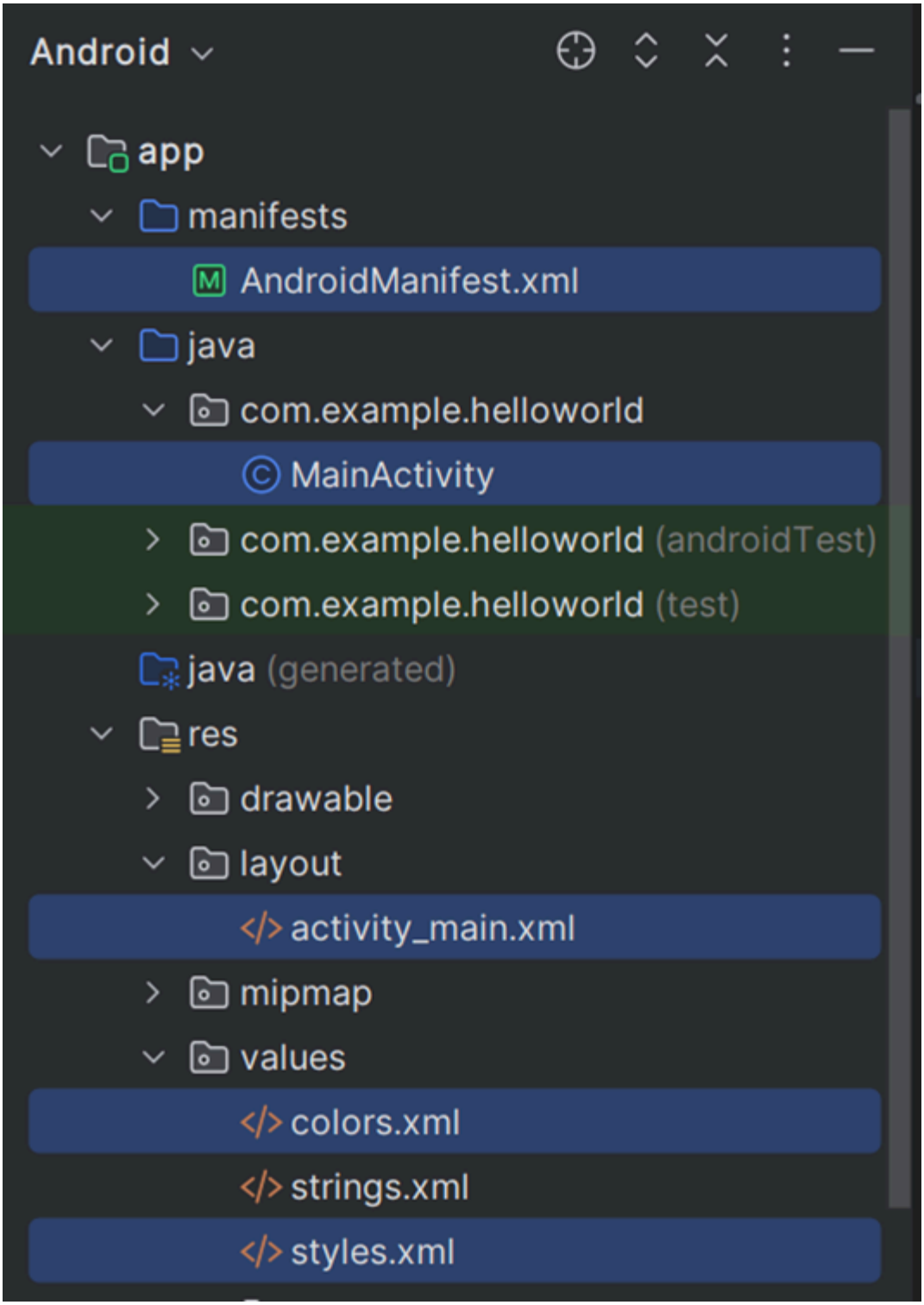
Project 1: Hello World

Project Name: HelloWorld

Template: Empty Views Activity

Project Architecture:

Total 5 files


Android
app
manifests
AndroidManifest.xml
java
com.example.helloworld
MainActivity
com.example.helloworld (androidTest)
com.example.helloworld (test)
java (generated)
res
drawable
layout
activity_main.xml
mipmap
values
colors.xml
strings.xml
styles.xml

I

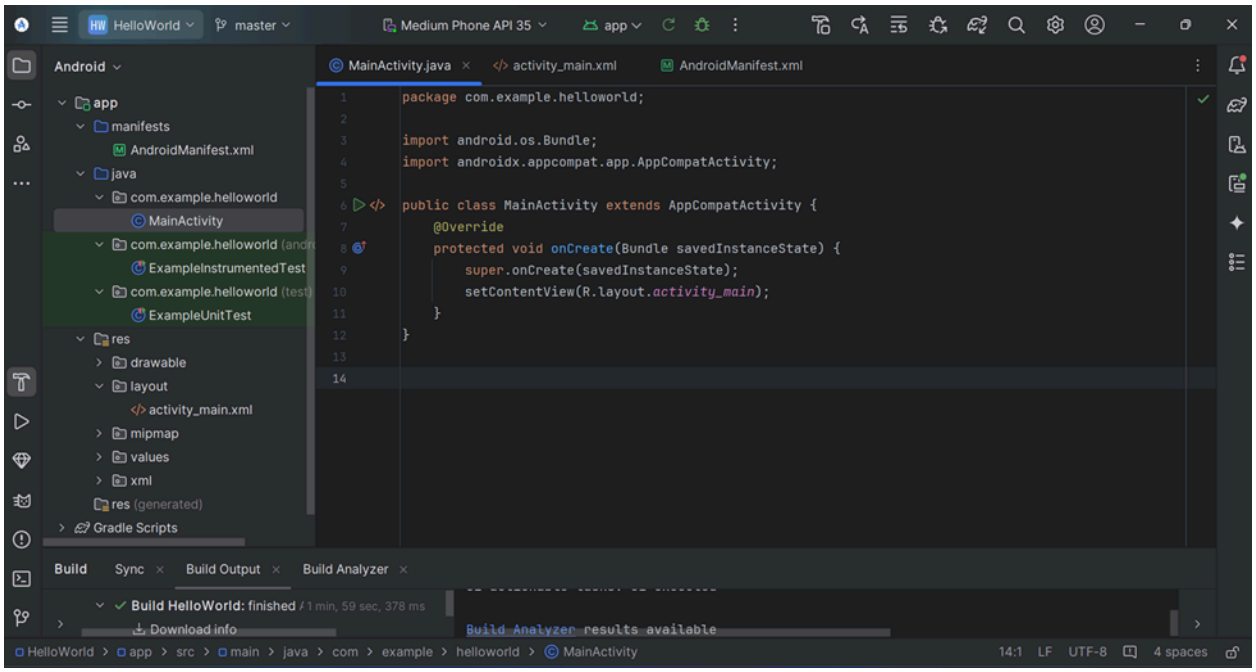
Filename: MainActivity.java

```
package com.example.helloworld;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {
      @Override
      protected void onCreate(Bundle savedInstanceState) {
         super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      }
}
```

Path: app>java>com.example.helloworld>MainActivity.java



II

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

        xmlns:android="http://schemas.android.com/apk/res/android"

        android:layout_width="match_parent"

        android:layout_height="match_parent"

        android:orientation="vertical"

        android:gravity="center"

        android:padding="16dp">


        <TextView

        android:id="@+id/text_hello"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Hello, World!"

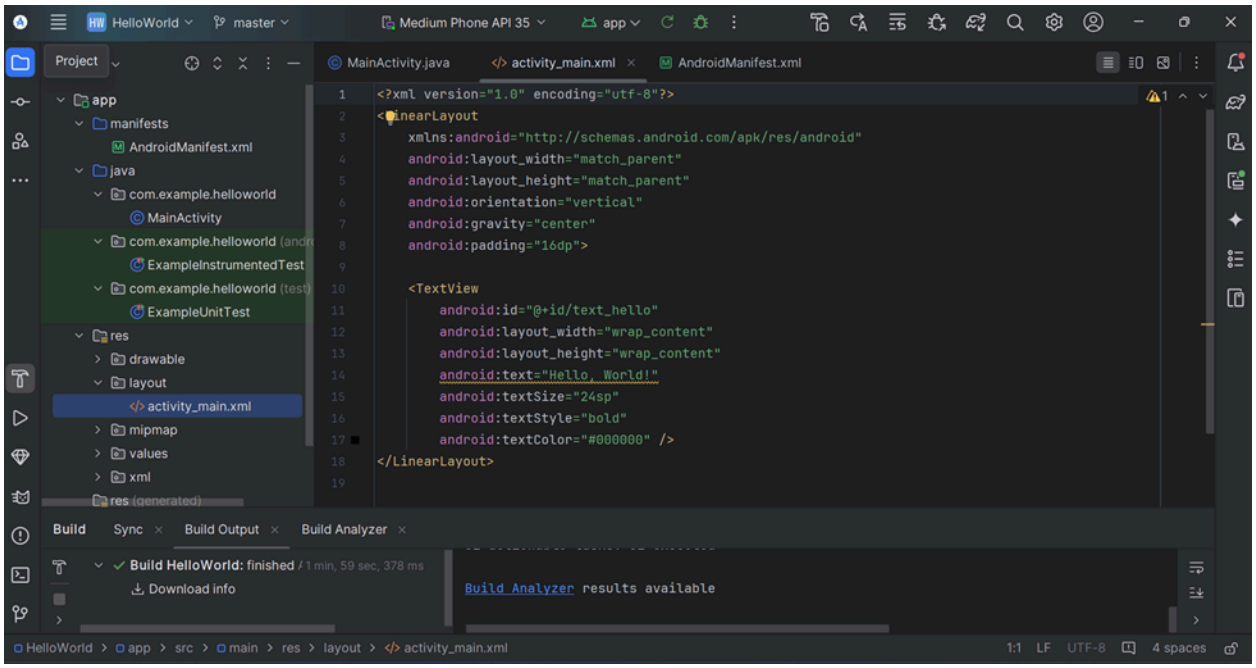
        android:textSize="24sp"

        android:textStyle="bold"

        android:textColor="#000000" />

</LinearLayout>
```

Path: app>res>layout>activity_main.xml



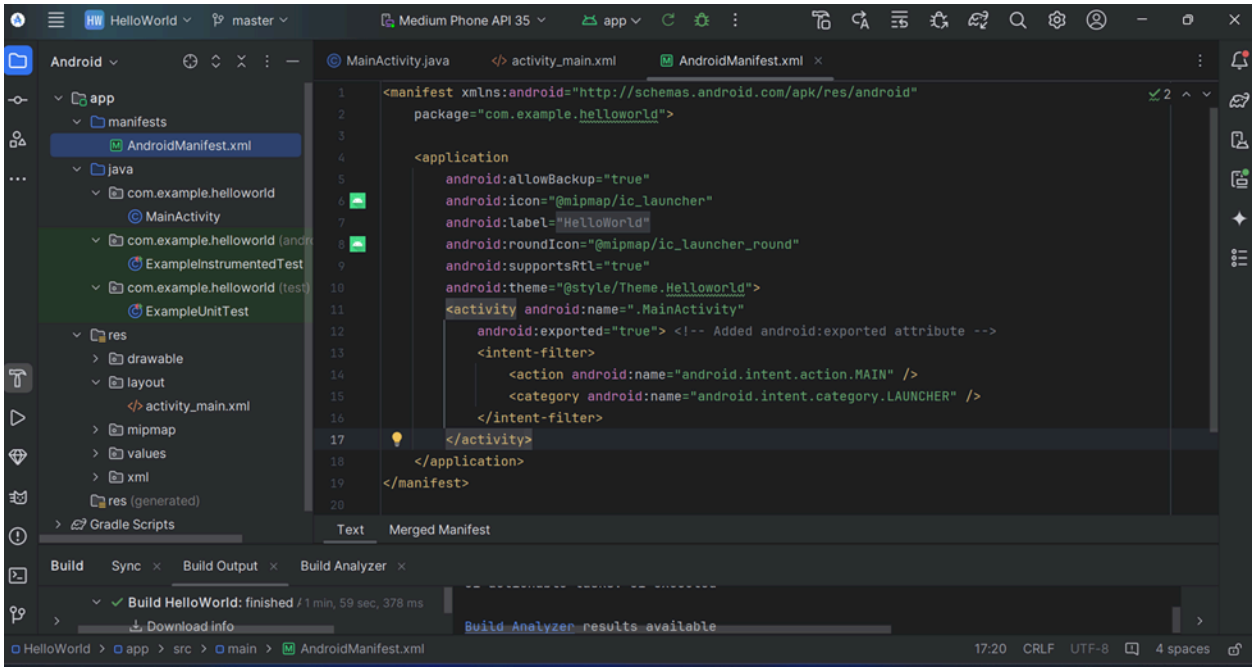
III

Filename: AndroidManifest.xml

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
        package="com.example.helloworld">
        <application
    android:allowBackup="true"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.Helloworld">
        <activity android:name=".MainActivity"
        android:exported="true"> <!-- Added android:exported attribute -->
        <intent-filter>
                <action android:name="android.intent.action.MAIN" />
          <category android:name="android.intent.category.LAUNCHER" />
        </intent-filter>
```

```
        </activity>
        </application>
</manifest>
```

Path: app>manifest>AndroidManifest.xml

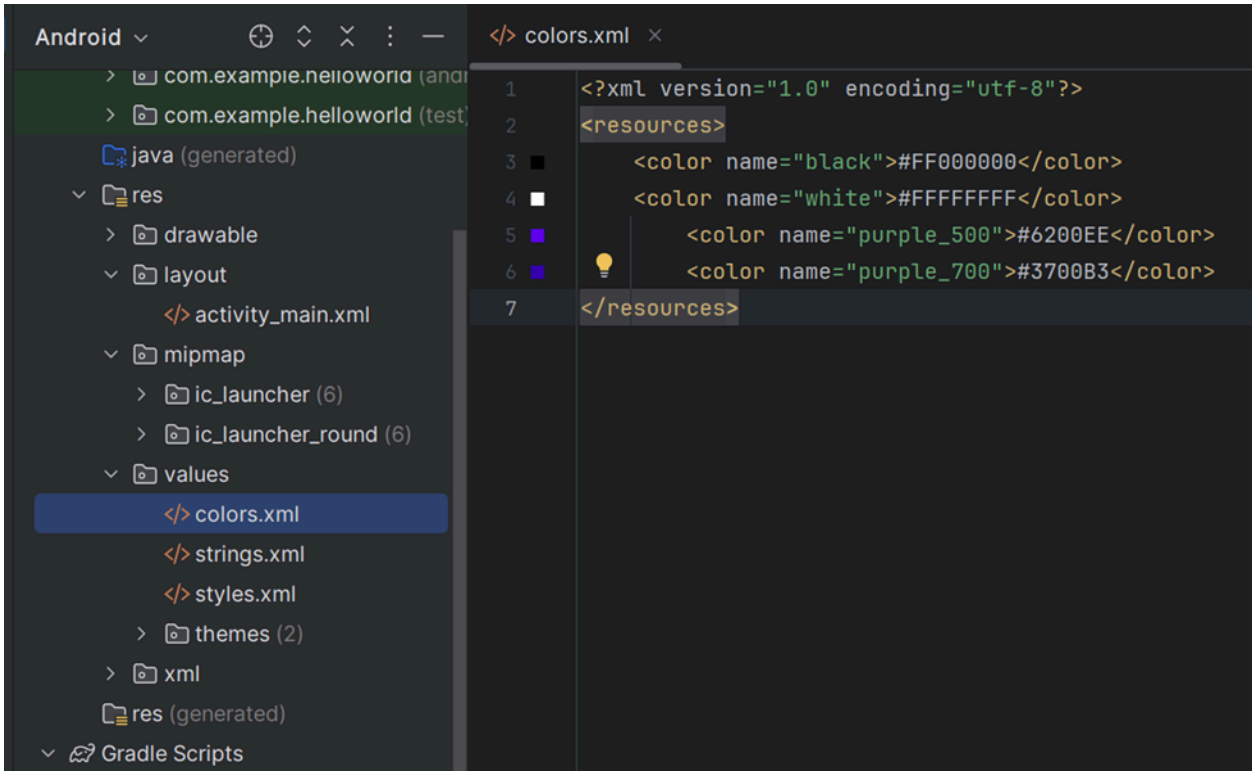


IV

Filename: colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <color name="black">#FF000000</color>
      <color name="white">#FFFFFFFF</color>
      <color name="purple_500">#6200EE</color>
      <color name="purple_700">#3700B3</color>
 </resources>
```

Path: app>res>values>colors.xml

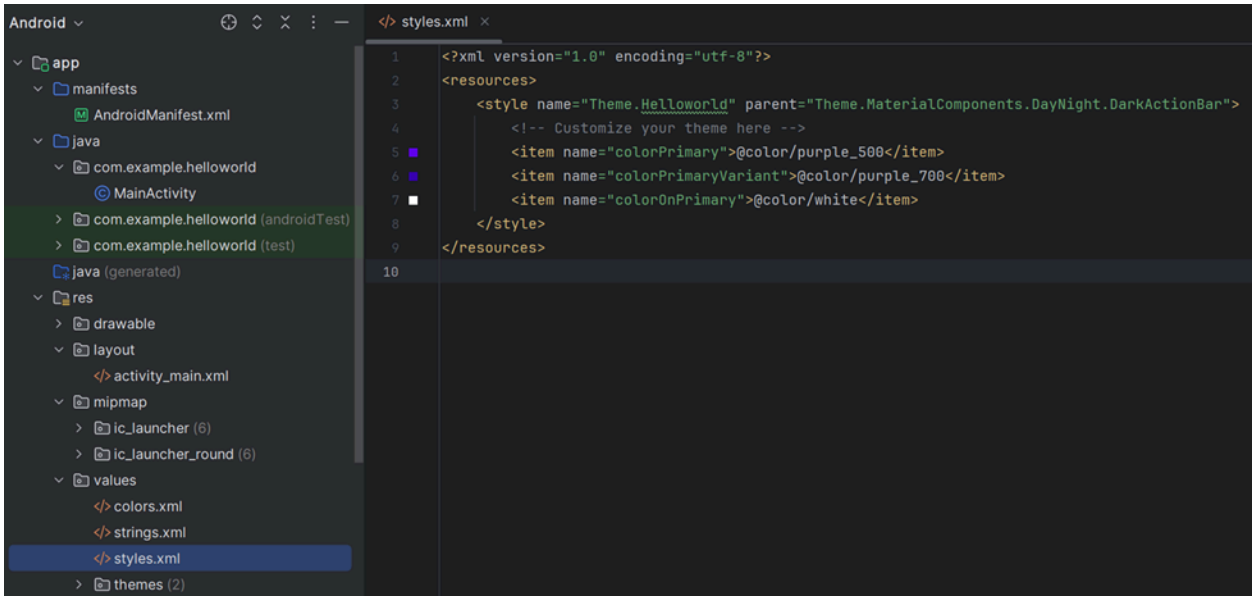


V

Filename: styles.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <style name="Theme.Helloworld"
parent="Theme.MaterialComponents.DayNight.DarkActionBar">
      <!-- Customize your theme here -->
      <item name="colorPrimary">@color/purple_500</item>
      <item name="colorPrimaryVariant">@color/purple_700</item>
      <item name="colorOnPrimary">@color/white</item>
      </style>
 </resources>
```

Path: app>res>values>styles.xml



VI (Prebuild Project Code: To ensure proper project detail: No need to update until needed)

Filename: build.gradle.kts (Module:app)

```
plugins {
      alias(libs.plugins.android.application)
 }

 android {
      namespace = "com.example.helloworld"
      compileSdk = 35

      defaultConfig {
      applicationId = "com.example.helloworld"
      minSdk = 24
      targetSdk = 35
      versionCode = 1
      versionName = "1.0"

      testInstrumentationRunner = "androidx.test.runner.AndroidJUnitRunner"
      }

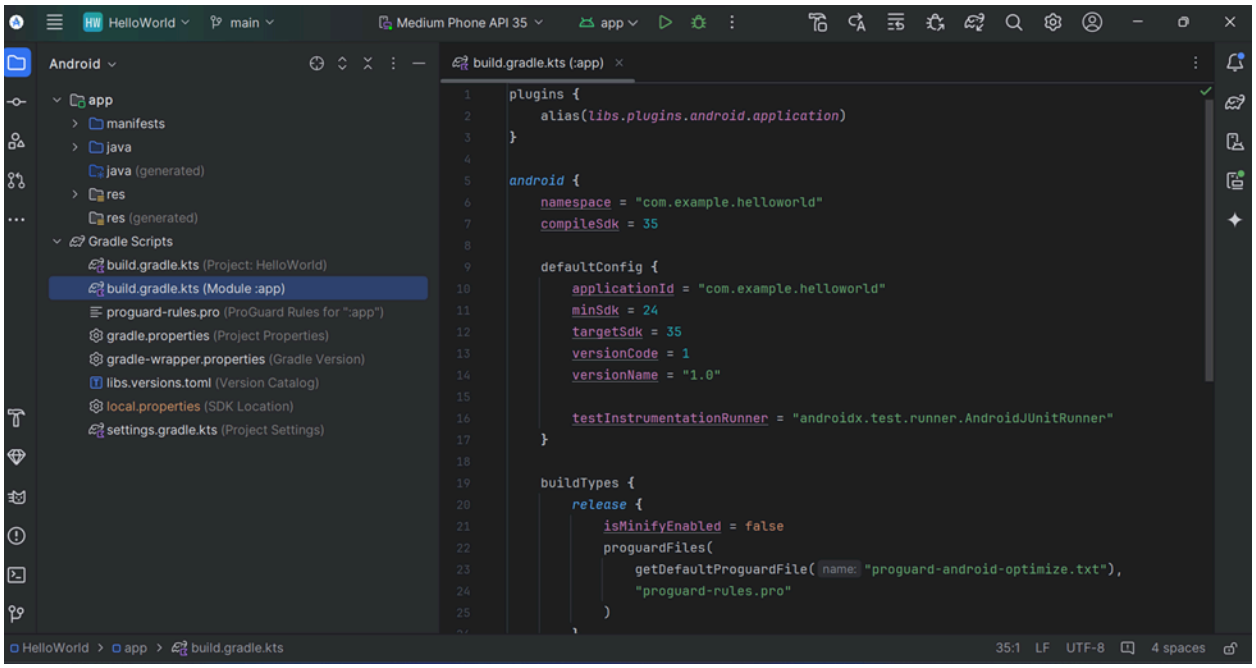
      buildTypes {
      release {
            isMinifyEnabled = false
            proguardFiles(
                  getDefaultProguardFile("proguard-android-optimize.txt"),
                  "proguard-rules.pro"
            )
      }
```

```
        }
        compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
        }
}

dependencies {

        implementation(libs.appcompat)
        implementation(libs.material)
        implementation(libs.activity)
        implementation(libs.constraintlayout)
        testImplementation(libs.junit)
        androidTestImplementation(libs.ext.junit)
        androidTestImplementation(libs.espresso.core)
}
```

Path: Gradle Scripts > build.gradle.kts

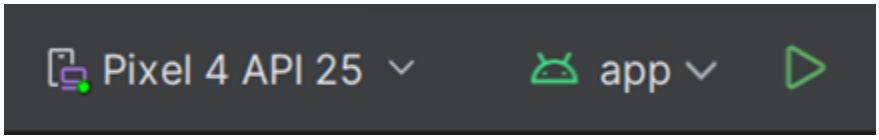


```
        }
    }
    compileOptions {
        sourceCompatibility = JavaVersion.VERSION_11
        targetCompatibility = JavaVersion.VERSION_11
    }
}

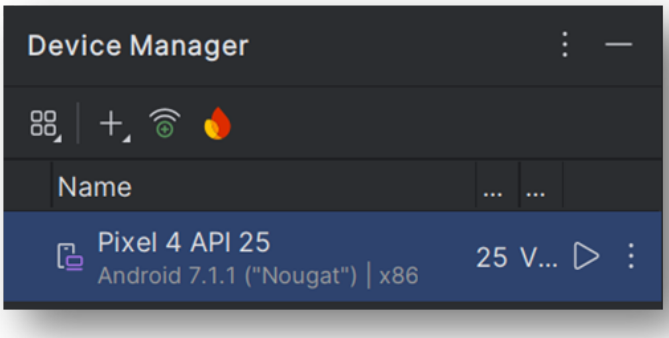
dependencies {

    implementation(libs.appcompat)
    implementation(libs.material)
    implementation(libs.activity)
    implementation(libs.constraintlayout)
    testImplementation(libs.junit)
    androidTestImplementation(libs.ext.junit)
    androidTestImplementation(libs.espresso.core)
}
```

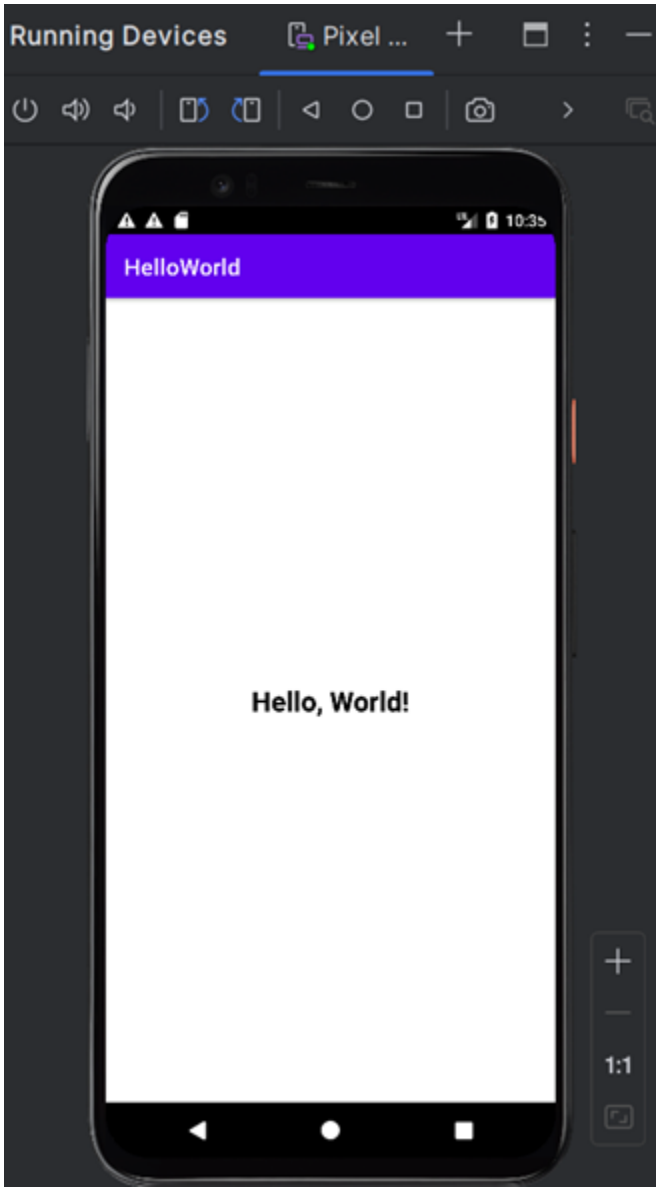
Above the Android Studio near Main Menu Click the Play button to Run the Program



Recommended compulsory AVD Emulator: Pixel 4 API 25 with version 7.1.1 Nougat



Output:



b.Develop an Android application demonstrating the use of Programming Resources (colors.xml, themes.xml, strings.xml, dimens.xml, drawable resources) and apply them in UI components.

Programming Resources

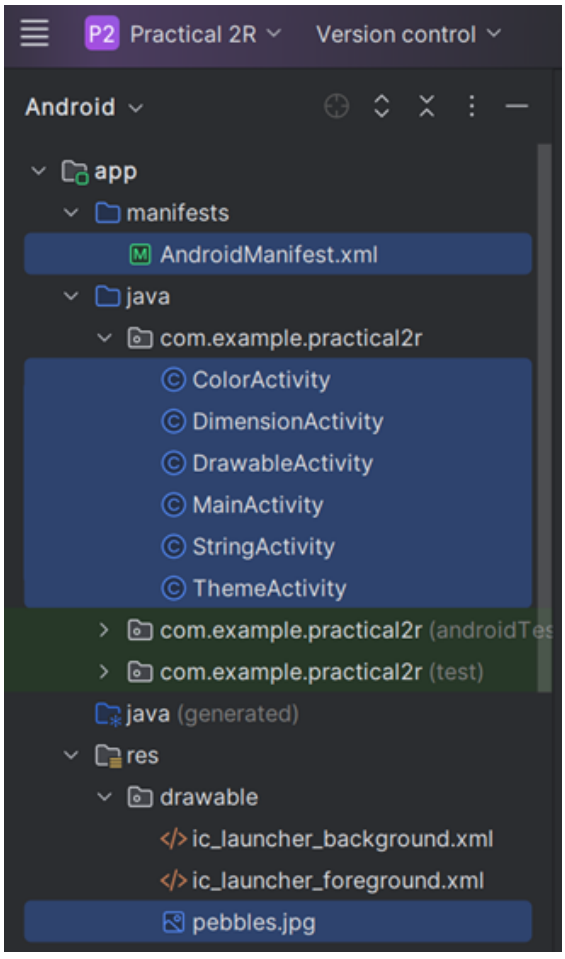
Android Resources: (Color, Theme, String, Drawable, Dimension, Image)

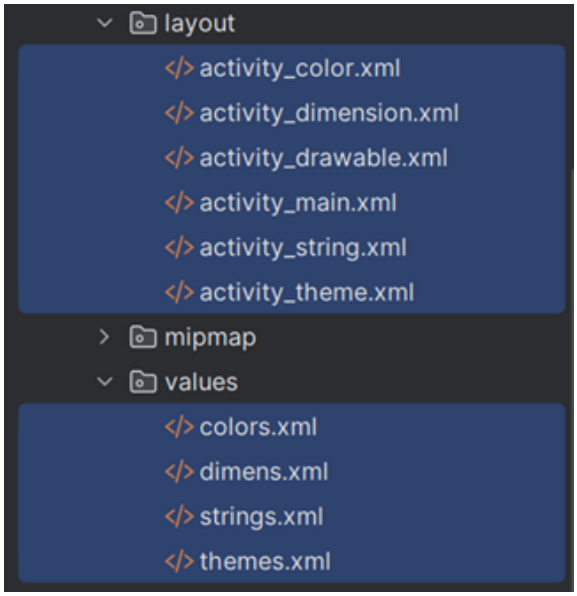
Project Name: Practical 2R

Template: Empty Views Activity

Language: Java

Project Architecture: 1 AndroidManifest.xml file, 6 Java Files,1 image in drawable image file 6 layout .xml files and 4 values .xml files: Total 18 files.





Filename: AndroidManifest.xml

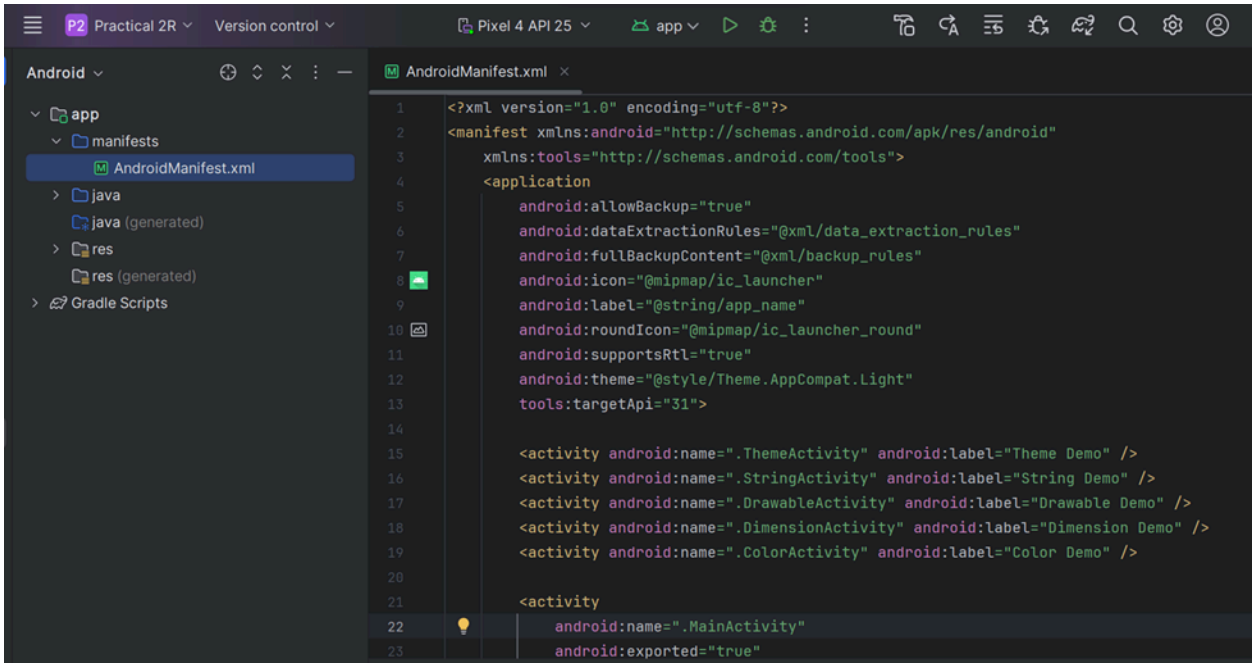
```
<?xml version="1.0" encoding="utf-8"?>
 <manifest xmlns:android="http://schemas.android.com/apk/res/android"
      xmlns:tools="http://schemas.android.com/tools">
      <application
      android:allowBackup="true"
      android:dataExtractionRules="@xml/data_extraction_rules"
      android:fullBackupContent="@xml/backup_rules"
      android:icon="@mipmap/ic_launcher"
      android:label="@string/app_name"
      android:roundIcon="@mipmap/ic_launcher_round"
      android:supportsRtl="true"
      android:theme="@style/Theme.AppCompat.Light"
      tools:targetApi="31">

      <activity android:name=".ThemeActivity" android:label="Theme Demo" />
      <activity android:name=".StringActivity" android:label="String Demo" />
      <activity android:name=".DrawableActivity" android:label="Drawable Demo"
/>
      <activity android:name=".DimensionActivity" android:label="Dimension
Demo" />
      <activity android:name=".ColorActivity" android:label="Color Demo" />

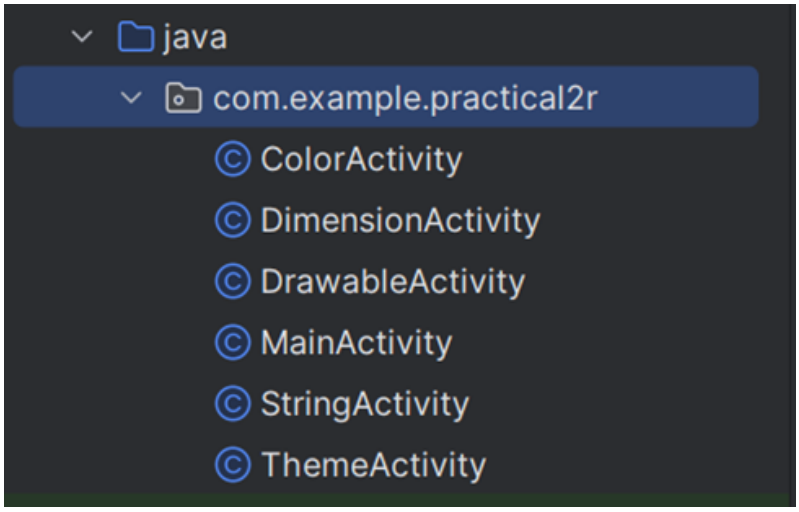
      <activity
            android:name=".MainActivity"
            android:exported="true"
            android:label="@string/app_name"
            android:theme="@style/Theme.AppCompat.Light">
            <intent-filter>
                  <action android:name="android.intent.action.MAIN" />
```

```
                <category android:name="android.intent.category.LAUNCHER" />
            </intent-filter>
        </activity>
        </application>
</manifest>
```

Path: app>manifests>AndroidManifest.xml



Total 6 .java files path



1.Filename: ColorActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class ColorActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_color);
     }
}
```

2.Filename: DimensionActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class DimensionActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_dimension);
     }
}
```

3.Filename: DrawableActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class DrawableActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_drawable);
     }
}
```

4.Filename: MainActivity.java

```
package com.example.practical2r;

 import android.content.Intent;
 import android.os.Bundle;
 import android.view.View;
 import androidx.appcompat.app.AppCompatActivity;

 public class MainActivity extends AppCompatActivity {
      @Override
      protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);
      }
      public void openColorActivity(View view) {
      startActivity(new Intent(this, ColorActivity.class));
      }
      public void openThemeActivity(View view) {
      startActivity(new Intent(this, ThemeActivity.class));
      }
      public void openStringActivity(View view) {
      startActivity(new Intent(this, StringActivity.class));
      }
      public void openDrawableActivity(View view) {
      startActivity(new Intent(this, DrawableActivity.class));
      }
      public void openDimensionActivity(View view) {
      startActivity(new Intent(this, DimensionActivity.class));
      }
 }
```

5.Filename: StringActivity.java

```
package com.example.practical2r;

 import android.os.Bundle;
 import androidx.appcompat.app.AppCompatActivity;

 public class StringActivity extends AppCompatActivity {
      @Override
      protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_string);
      }
 }
```

6.Filename: ThemeActivity.java

```
package com.example.practical2r;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class ThemeActivity extends AppCompatActivity {
     @Override
     protected void onCreate(Bundle savedInstanceState) {
     super.onCreate(savedInstanceState);
     setContentView(R.layout.activity_theme);
     }
}
```

Image File Path

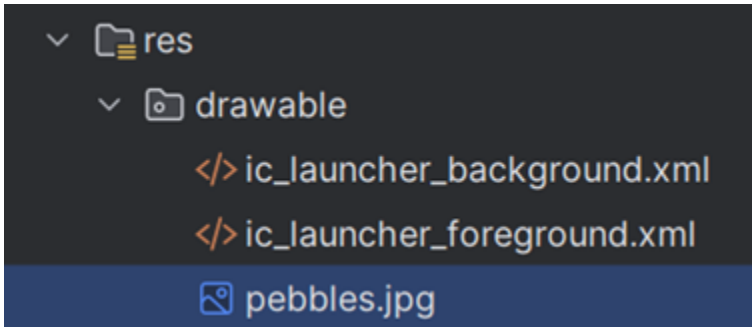
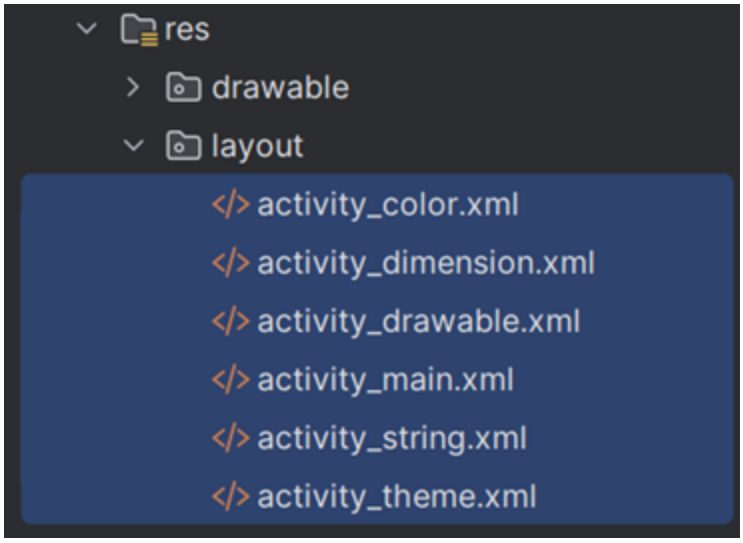


Image Detail: pebbles.jpg

(Copy your image file from your local folder and only copy and paste into drawable folder. Don't Drag and Drop). Image size must within KB size



Total 6 Layout .xml files path



1.Filename: activity_color.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:background="@color/colorRed">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Color Activity"
      android:textSize="24sp"
          android:layout_centerInParent="true"
      android:textColor="@color/white"/>
 </RelativeLayout>
```

2.Filename: activity_dimension.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical"
      android:padding="16dp"
      android:gravity="center">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Dimension Activity"
      android:textSize="24sp"
```

```
      android:textColor="@android:color/black"/>
      <!-- Button demonstrating dimension usage -->
      <Button
         android:id="@+id/dimensionButton"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Click Me!"
      android:textSize="@dimen/button_text_size"
      android:padding="@dimen/button_padding"
      android:layout_marginTop="@dimen/button_margin_top"
      android:background="@color/btn_color"/>
 </LinearLayout>
```

3.Filename: activity_drawable.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:background="@drawable/pebbles">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Drawable Activity"
      android:textSize="24sp"
      android:layout_centerInParent="true"
      android:textColor="@android:color/white"/>
 </RelativeLayout>
```

4.Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout
      xmlns:android="http://schemas.android.com/apk/res/android"
      xmlns:tools="http://schemas.android.com/tools"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      tools:context=".MainActivity"
      android:background="@drawable/pebbles">
      <Button
      android:id="@+id/btnColor"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Color Demo"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="50dp"
      android:onClick="openColorActivity"/>
      <Button
      android:id="@+id/btnTheme"
```

```
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Theme Demo"
      android:layout_below="@id/btnColor"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
      android:onClick="openThemeActivity"/>
      <Button
      android:id="@+id/btnString"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="String Demo"
      android:layout_below="@id/btnTheme"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
         android:onClick="openStringActivity"/>

      <Button
      android:id="@+id/btnDrawable"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Drawable Demo"
      android:layout_below="@id/btnString"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
      android:onClick="openDrawableActivity"/>
      <Button
      android:id="@+id/btnDimension"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="Dimension Demo"
      android:layout_below="@id/btnDrawable"
      android:layout_centerHorizontal="true"
      android:layout_marginTop="20dp"
      android:onClick="openDimensionActivity"/>
</RelativeLayout>
```

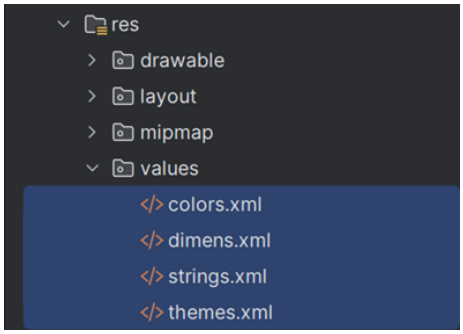
5.Filename: activity_string.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:orientation="vertical"
      android:gravity="center"
      android:background="@color/colorPrimaryDark">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="@string/Welcome_Message"
      android:textSize="24sp"
      android:textColor="@android:color/white"/>
 </LinearLayout>
```

6.Filename: activity_theme.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
      android:layout_width="match_parent"
      android:layout_height="match_parent"
      android:background="@android:color/holo_blue_bright">
      <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="You got this sky blue theme"
      android:textSize="24sp"
      android:layout_centerInParent="true"
      android:textColor="@android:color/black"/>
 </RelativeLayout>
```

Total 4 values .xml files



1.Filename: colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <color name="purple_200">#FFBB86FC</color>
      <color name="purple_500">#FF6200EE</color>
      <color name="purple_700">#FF3700B3</color>
      <color name="teal_200">#9ADA03</color>
      <color name="teal_700">#FF018786</color>
      <color name="black">#FF000000</color>
      <color name="white">#FFFFFFFF</color>
      <color name="btn_color">#FFC49662</color>
      <color name="colorRed">#E91E63</color>
      <color name="colorBlue">#0000FF</color>
      <color name="colorPrimary">#6200EE</color>
      <color name="colorPrimaryDark">#3700B3</color>
      <color name="colorAccent">#03DAC5</color>
 </resources>
```

2.Filename: dimens.xml

```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <dimen name="button_margin_top">24dp</dimen>
      <dimen name="button_padding">16dp</dimen>
      <dimen name="button_text_size">18sp</dimen>
 </resources>
```

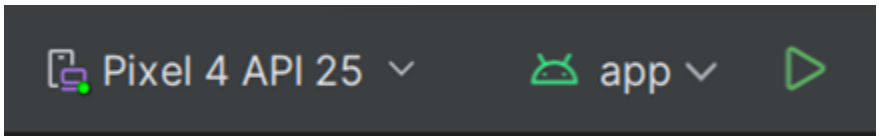
3.Filename: strings.xml

```
<resources>
      <string name="app_name">Practical 2R</string>
      <string name="button_name">Click on me</string>
      <string name="ParaHead">Welcome to Practical 2R</string>
      <string name="Description">This is a string resource demo.</string>
      <string name="Welcome_Message">Welcome to Android Studio. This is your
Practical 2 Resources.</string>
 </resources>
```

4.Filename: themes.xml

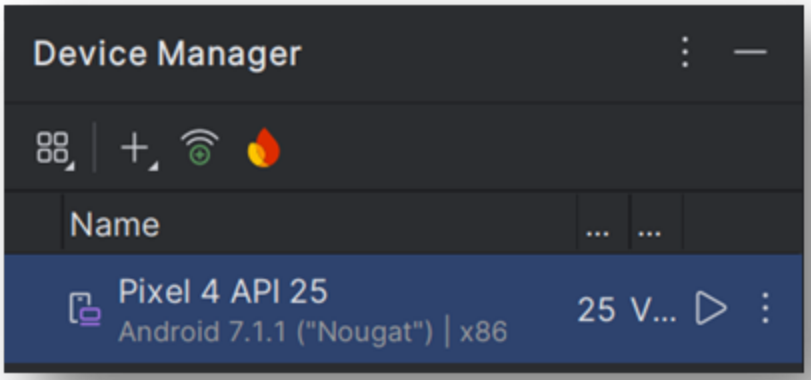
```
<?xml version="1.0" encoding="utf-8"?>
 <resources>
      <style name="AppTheme" parent="Theme.AppCompat.Light.DarkActionBar">
        <item name="colorPrimary">@color/colorPrimary</item>
      <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
      <item name="colorAccent">@color/colorAccent</item>
      </style>
      <style name="BlueTheme" parent="Theme.AppCompat.Light.DarkActionBar">
      <item name="colorPrimary">@color/colorPrimary</item>
      <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
      <item name="colorAccent">@color/colorAccent</item>
      <item name="android:background">#00F0EF</item>
      <item name="android:textColor">#000000</item>
      </style>
 </resources>
```

Above the Android Studio near Main Menu Click the Play button to Run the Program



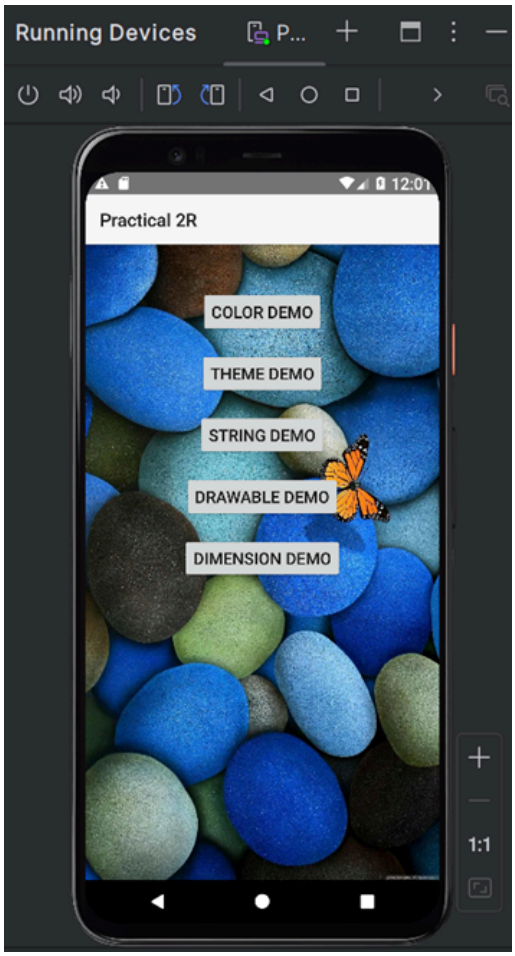
Recommended compulsory AVD Emulator: Pixel 4 API 25 with version 7.1.1 Nougat

Play Emulator

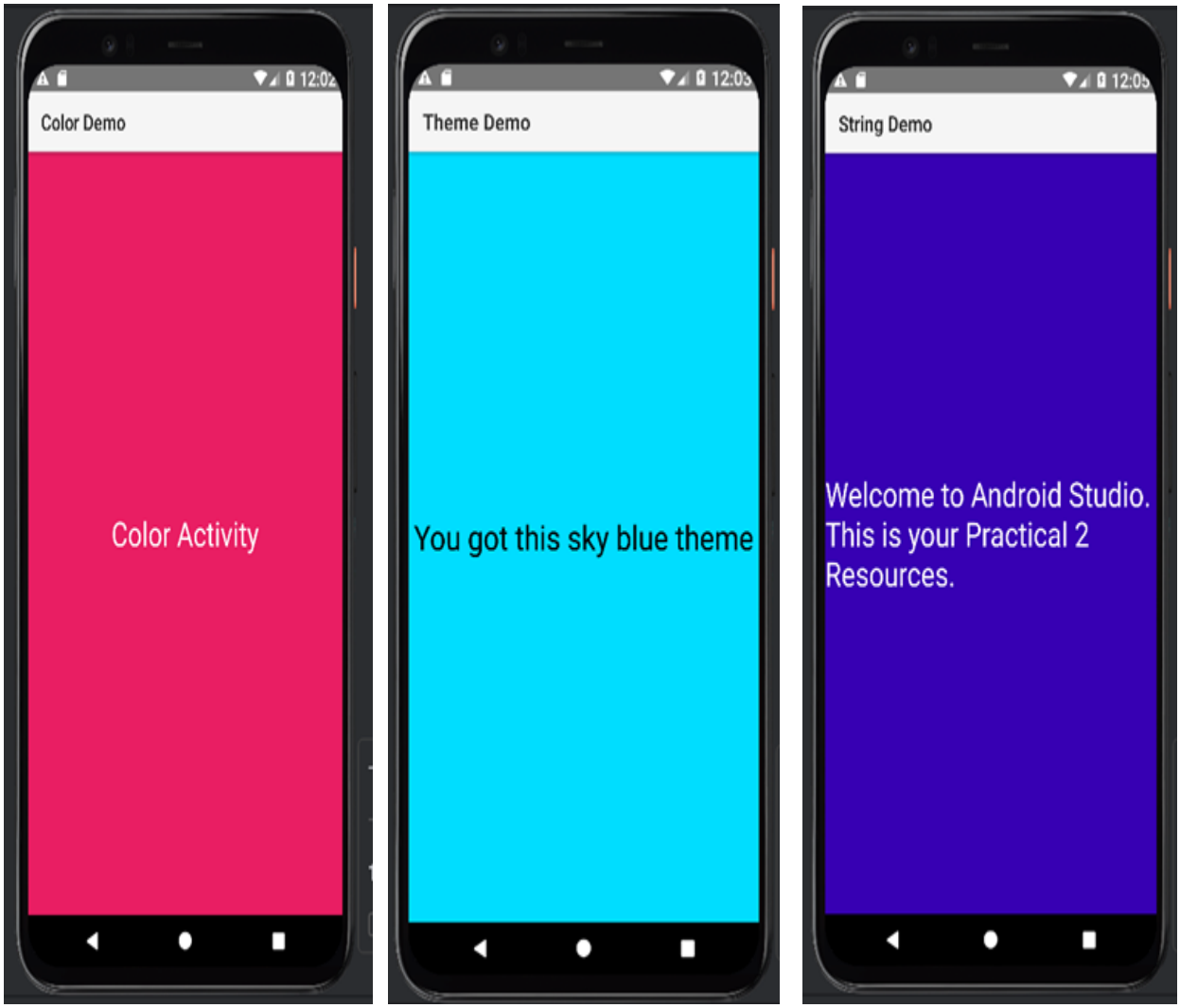


Output:

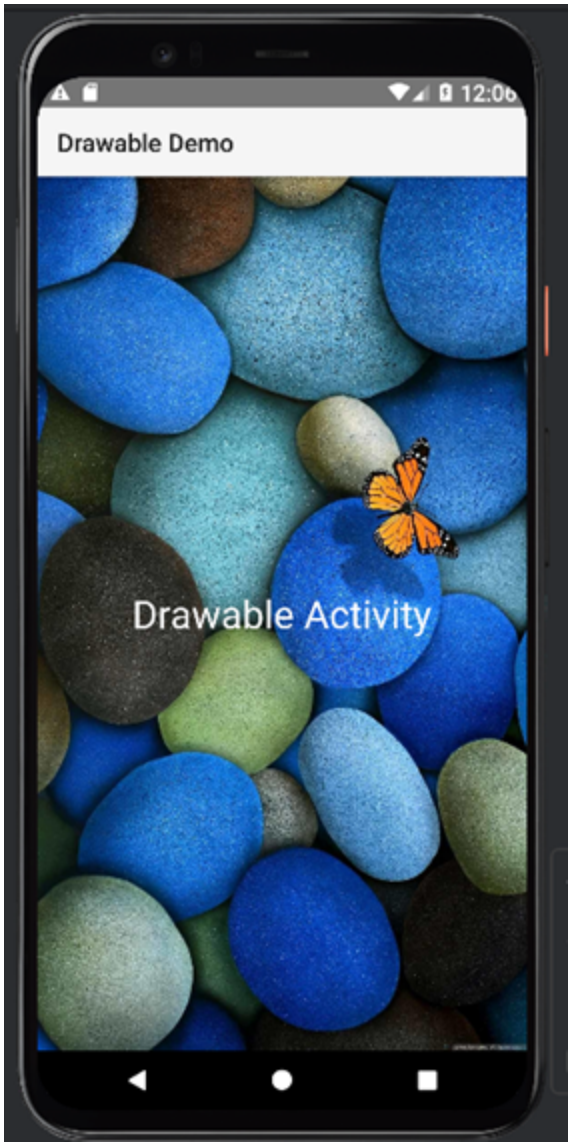
MAIN PAGE



Click COLOR DEMO press back & Click THEME DEMO press back & Click THEME DEMO



Press back & Click COLOR DEMO Press back & Click COLOR DEMO


12:06
Drawable Demo
Drawable Activity


12:06
Dimension Demo
Dimension Activity
CLICK ME!

C. Develop an Android application demonstrating complete Activity Lifecycle methods and analyze execution using Logcat.

Project Name: logcat

```
package com.example.logact;

import android.os.Bundle;
import android.util.Log;

import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {



    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d("on start invoked", "on started");

    }

    @Override
    protected void onPostResume() {
        super.onPostResume();
        Log.d("on resume invoked", "on resume");
    }

    @Override
    protected void onPause() {
        super.onPause();
        Log.d("on pause invoked", "on pause");
    }

    @Override
        protected void onStop() {
```

```
            super.onStop();
            Log.d("on stop invoked", "on stopped");
        }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d("on destroy invoked", "on destroy");
    }
}
```

2. Fragments and Activity Interaction

4. Develop an Android application to create and integrate a Fragment and demonstrate Fragment lifecycle methods.

Project Name: FragmentLifecycleDemo

Create a new fragment class:

java->new-> fragment->fragment(blank)

Fragment Name: DemoFragment

Fragment Layout name: fragment_demo

I

Filename: DemoFragment.java

```
package com.example.fragmentlifecycledemo;

import android.content.Context;

import android.os.Bundle;

import android.util.Log;

import android.view.LayoutInflater; import android.view.View;

import android.view.ViewGroup;

import androidx.annotation.NonNull; import androidx.annotation.Nullable;

import androidx.fragment.app.Fragment;

public class DemoFragment extends Fragment {

   String TAG = "FragmentLifecycle";


   @Override

   public void onAttach(@NonNull Context context) {

   super.onAttach(context);
```

```
   Log.d(TAG, "onAttach() executed");}



   @Override

  public void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       Log.d(TAG, "onCreate() executed");

   }



   @Override

   public View onCreateView(LayoutInflater inflater, ViewGroup
container,Bundle savedInstanceState) {

       Log.d(TAG, "onCreateView() executed");

       return inflater.inflate(R.layout.fragment_demo, container, false);

   }



@Override

public void onViewCreated(@NonNull View view, @Nullable Bundle
savedInstanceState) {

       super.onViewCreated(view, savedInstanceState);

        Log.d("on view created invoked", "onViewCreated() executed");

}



   @Override

   public void onStart() {

       super.onStart();

       Log.d("on start invoked", "on started");
```

```
}


@Override
public void onResume() {
    super.onResume();
    Log.d("on resume invoked", "on resume");
}


@Override
public void onPause() {
    super.onPause();
    Log.d("on pause invoked", "on pause");
}


@Override
public void onStop() {
    super.onStop();
    Log.d("on stop invoked", "on stopped");
}


@Override
public void onDestroyView() {
    super.onDestroyView();
    Log.d("on destroyview invoked", "on destroyview() executed");
}
```

```
    @Override
    public void onDestroy() {
        super.onDestroy();
        Log.d("on destroy invoked", "on destroy");
    }

    @Override
    public void onDetach() {
        super.onDetach();
        Log.d("on detach invoked", "on detach");
    }
}
```

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/fragment_container_view_tag"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
```

```
        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Hello World!"

        app:layout_constraintBottom_toBottomOf="parent"

        app:layout_constraintEnd_toEndOf="parent"

        app:layout_constraintStart_toStartOf="parent"

        app:layout_constraintTop_toTopOf="parent" />


</androidx.constraintlayout.widget.ConstraintLayout>
```

Filename: MainActivity.java

```
package com.example.fragmentlifecycledemo;


import android.os.Bundle;

import androidx.fragment.app.FragmentTransaction;

import androidx.activity.EdgeToEdge;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.graphics.Insets;

import androidx.core.view.ViewCompat;

import androidx.core.view.WindowInsetsCompat;


public class MainActivity extends AppCompatActivity {


    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);

                DemoFragment fragment = new DemoFragment();

                FragmentTransaction transaction =

                        getSupportFragmentManager().beginTransaction();

                transaction.replace(R.id.fragment_container_view_tag,
fragment);

                transaction.commit();

        }

    }
```

fragment_demo.xml

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".DemoFragment">



    <!-- TODO: Update blank fragment layout -->

    <TextView

        android:layout_width="match_parent"

        android:layout_height="match_parent"

        android:text="@string/hello_blank_fragment" />



</FrameLayout>
```

5. Develop an Android application implementing Fragment-based UI navigation between multiple screens.

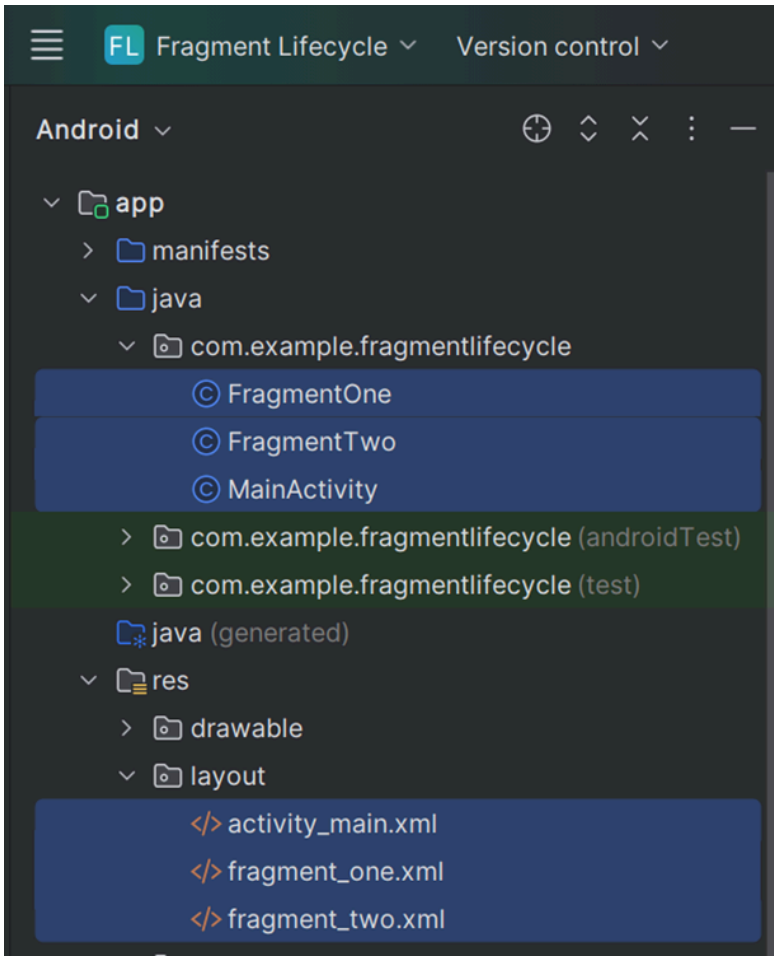
Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.

Project Name: Fragment Lifecycle

Template: Empty Views Activity

Language: Java

Project Architecture: Total 6 files



I Filename: MainActivity.java

```
package com.example.fragmentlifecycle;

 import androidx.appcompat.app.AppCompatActivity;
 import android.os.Bundle;
 import android.widget.Button;

 public class MainActivity extends AppCompatActivity {

      private boolean isFragmentOneLoaded = true;
      private final androidx.fragment.app.FragmentManager fragmentManager =
getSupportFragmentManager();

      @Override
      protected void onCreate(Bundle savedInstanceState) {
      super.onCreate(savedInstanceState);
      setContentView(R.layout.activity_main);

        Button btnChange = findViewById(R.id.btn_change);  // Ensure ID
matches activity_main.xml
      loadFragmentOne();

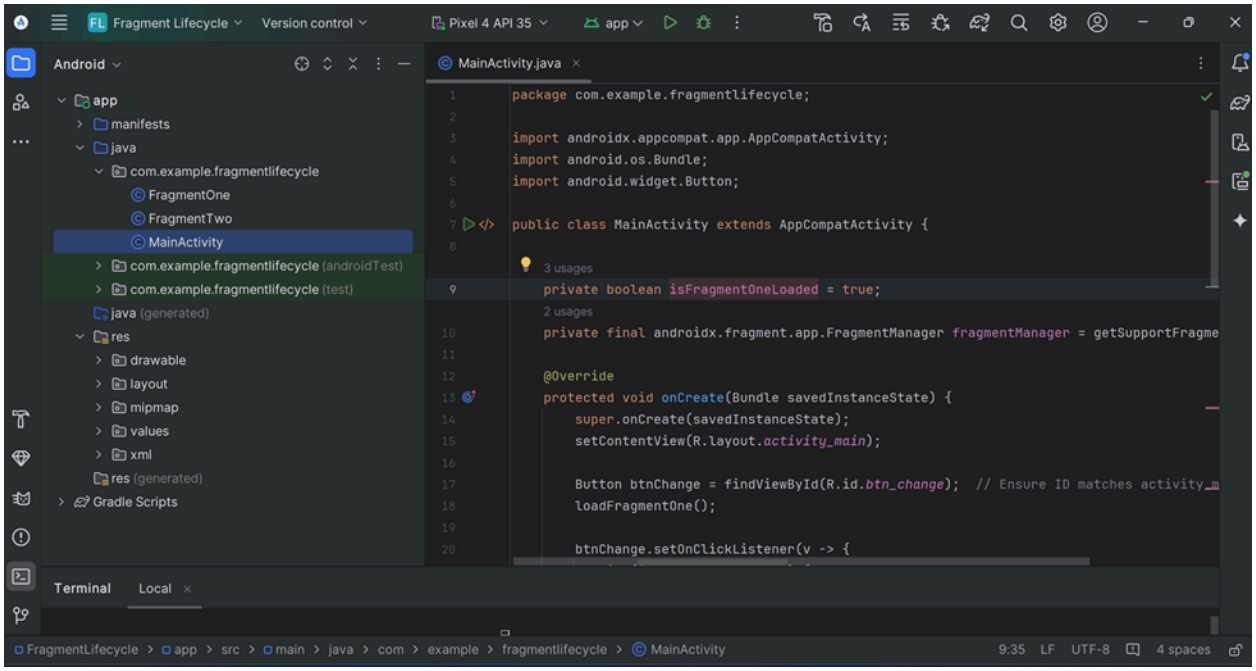
      btnChange.setOnClickListener(v -> {
            if (isFragmentOneLoaded) {
                  loadFragmentTwo();
            } else {
                  loadFragmentOne();
            }
      });
      }

      private void loadFragmentOne() {
         androidx.fragment.app.FragmentTransaction transaction =
fragmentManager.beginTransaction();
      FragmentOne fragment = new FragmentOne();
      transaction.replace(R.id.fragment_holder, fragment);  // Ensure ID
matches activity_main.xml
      transaction.addToBackStack(null);
      transaction.commit();
      isFragmentOneLoaded = true;
      }

      private void loadFragmentTwo() {
         androidx.fragment.app.FragmentTransaction transaction =
fragmentManager.beginTransaction();
      FragmentTwo fragment = new FragmentTwo();
      transaction.replace(R.id.fragment_holder, fragment);  // Ensure ID
matches activity_main.xml
```

```
        transaction.addToBackStack(null);
        transaction.commit();
        isFragmentOneLoaded = false;
        }
}
```

Path: app>java>com.example.fragmentlifecycle>MainActivity.java



II

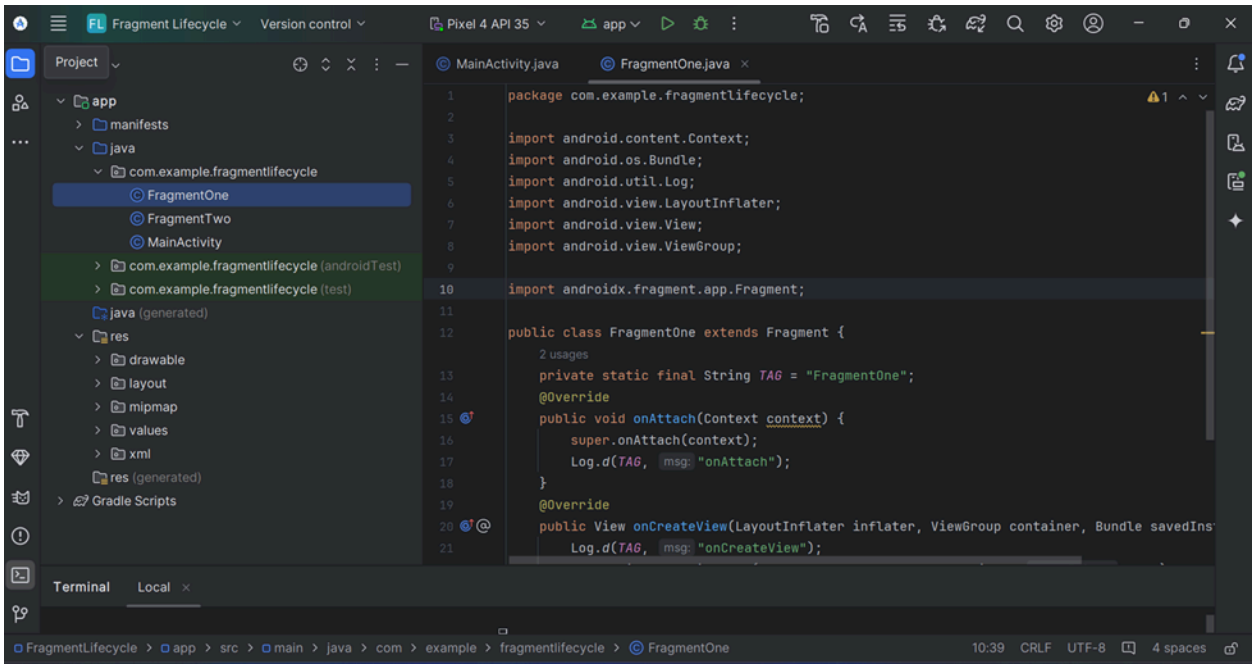
Filename: FragmentOne.java

```
package com.example.fragmentlifecycle;
import android.content.Context;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment;
public class FragmentOne extends Fragment {
        private static final String TAG = "FragmentOne";
        @Override
        public void onAttach(Context context) {
        super.onAttach(context);
```

```
        Log.d(TAG, "onAttach");
        }
        @Override
        public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
savedInstanceState) {
        Log.d(TAG, "onCreateView");
        return inflater.inflate(R.layout.fragment_one, container, false);
        }

}
```

Path: app>java>com.example.fragmentlifecycle > FragmentOne.java



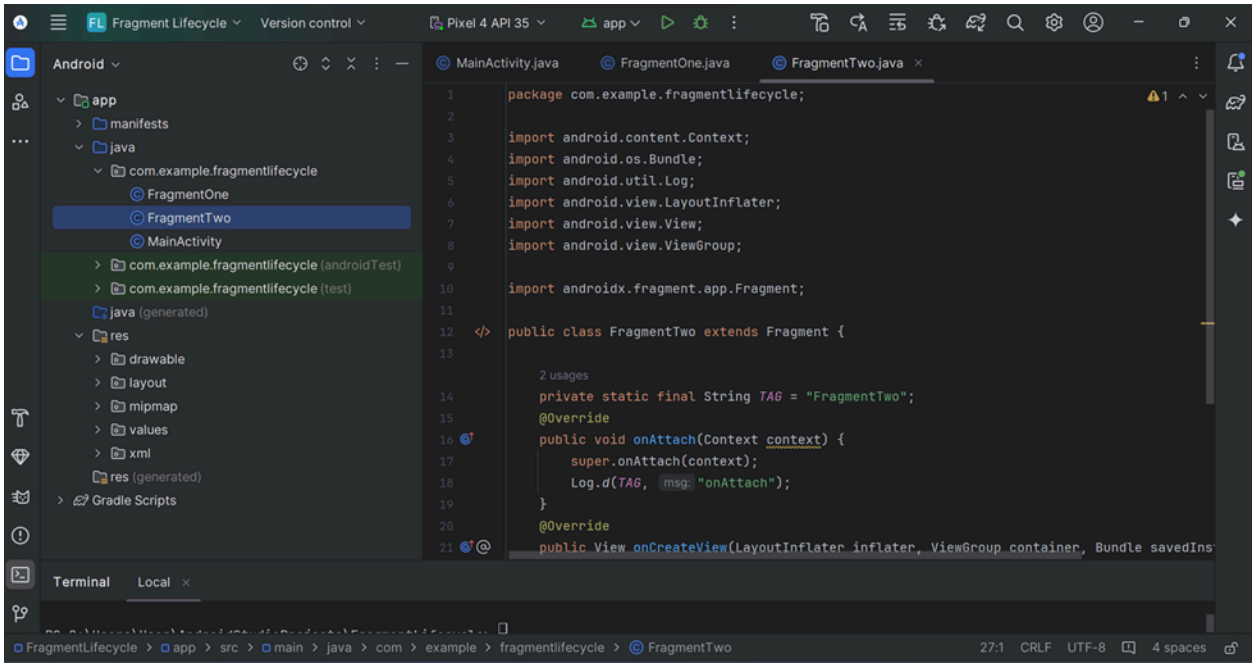
III

Filename: FragmentTwo.java

```
package com.example.fragmentlifecycle;
import android.content.Context;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment;
public class FragmentTwo extends Fragment {
        private static final String TAG = "FragmentTwo";
```

```
        @Override
        public void onAttach(Context context) {
        super.onAttach(context);
      Log.d(TAG, "onAttach");
        }\
        @Override
        public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle
savedInstanceState) {
        Log.d(TAG, "onCreateView");
        return inflater.inflate(R.layout.fragment_two, container, false);
        }
}
```

Path: app>java>com.example.fragmentlifecycle > FragmentTwo.java



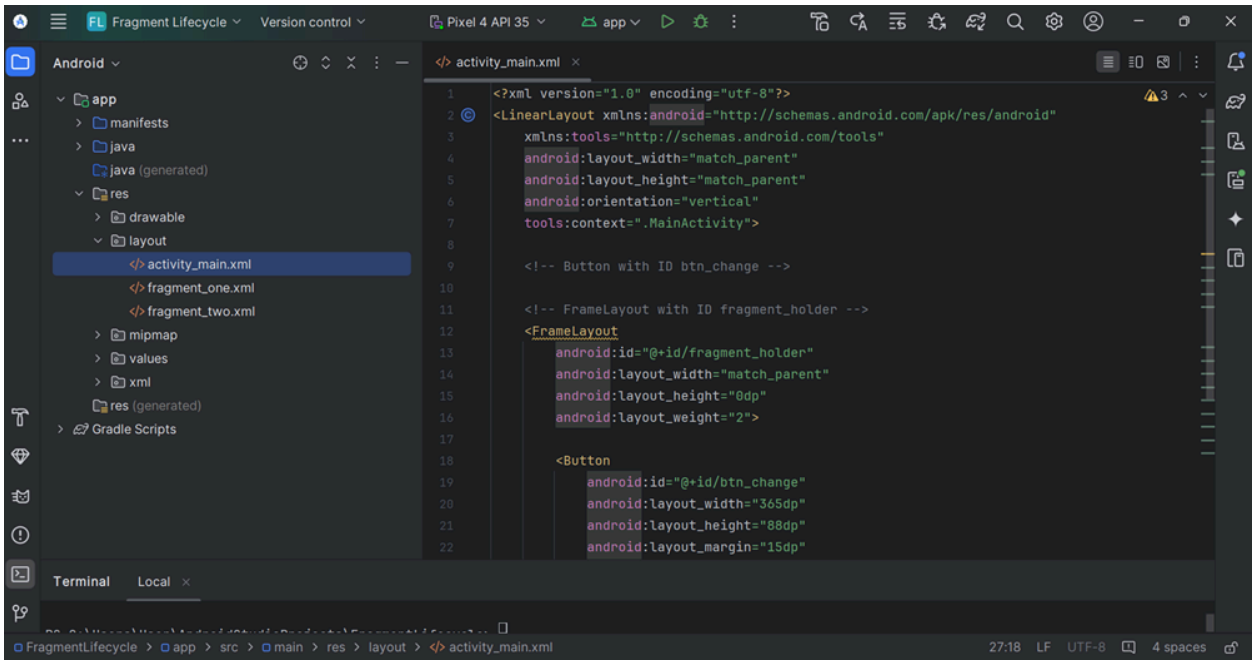
IV

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        xmlns:tools="http://schemas.android.com/tools"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        tools:context=".MainActivity">
```

```
        <!-- Button with ID btn_change -->
        <!-- FrameLayout with ID fragment_holder -->
        <FrameLayout
        android:id="@+id/fragment_holder"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="2">
        <Button
        android:id="@+id/btn_change"
        android:layout_width="365dp"
        android:layout_height="88dp"
        android:layout_margin="15dp"
        android:layout_marginTop="15dp"
        android:backgroundTint="@android:color/holo_green_dark"
        android:text="Change Fragment"
        android:textAllCaps="false"
        android:textSize="24sp" />
        </FrameLayout>
</LinearLayout>
```

Path: app>res>layout>activity_main.xml

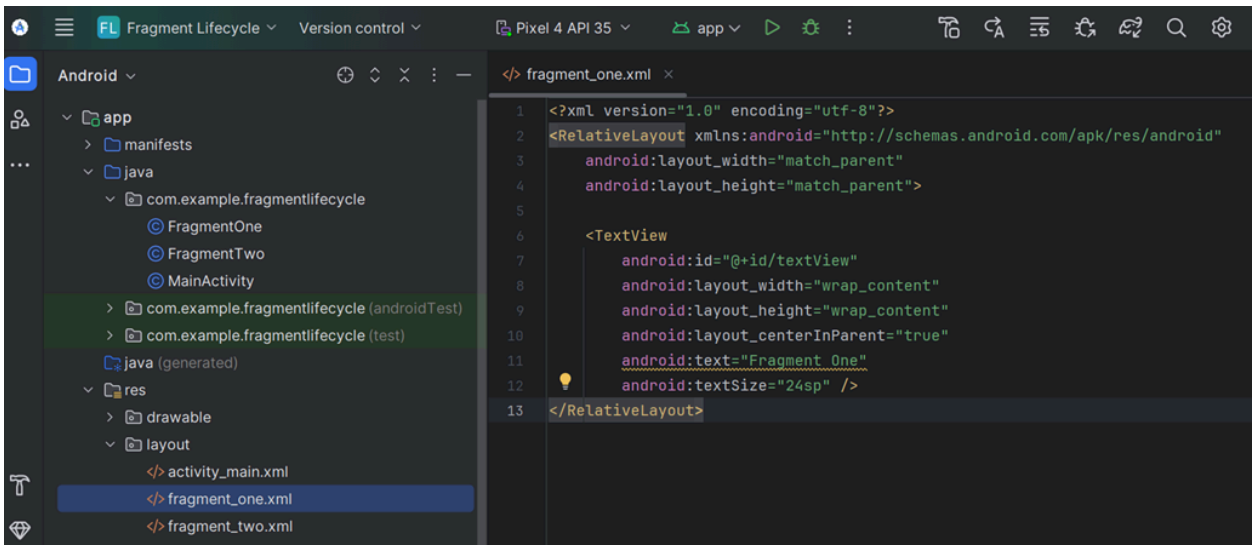


V

Filename: fragment_one.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
        <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Fragment One"
        android:textSize="24sp" />
</RelativeLayout>
```

Path: app>res>layout>fragment_one.xml



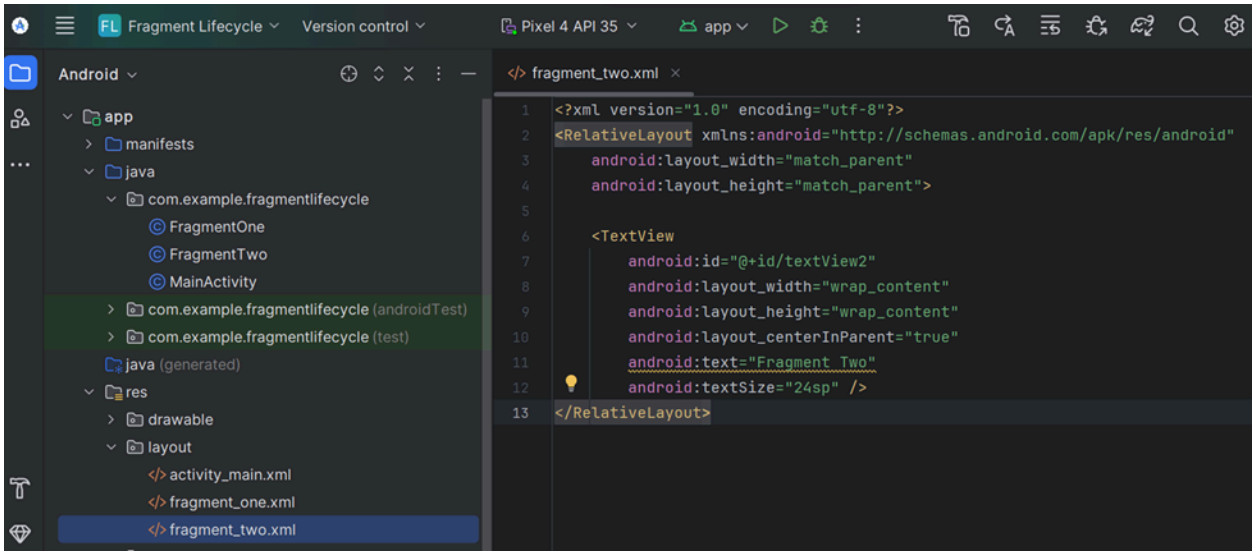
VI

Filename: fragment_two.xml

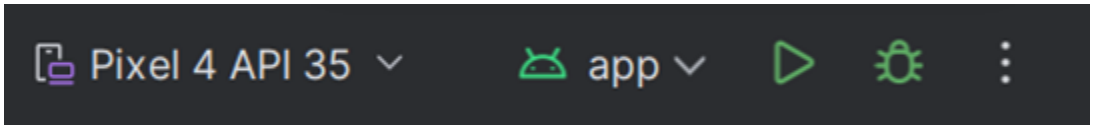
```
<?xml version="1.0" encoding="utf-8"?>
```

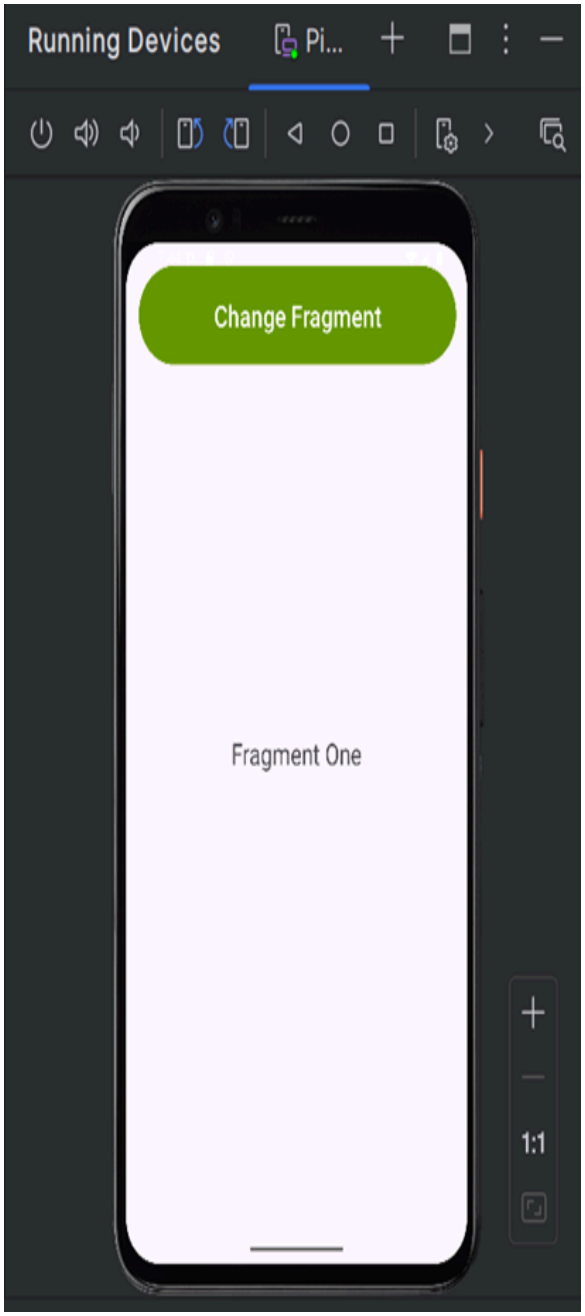
```
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
        <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Fragment Two"
        android:textSize="24sp" />
</RelativeLayout>
```

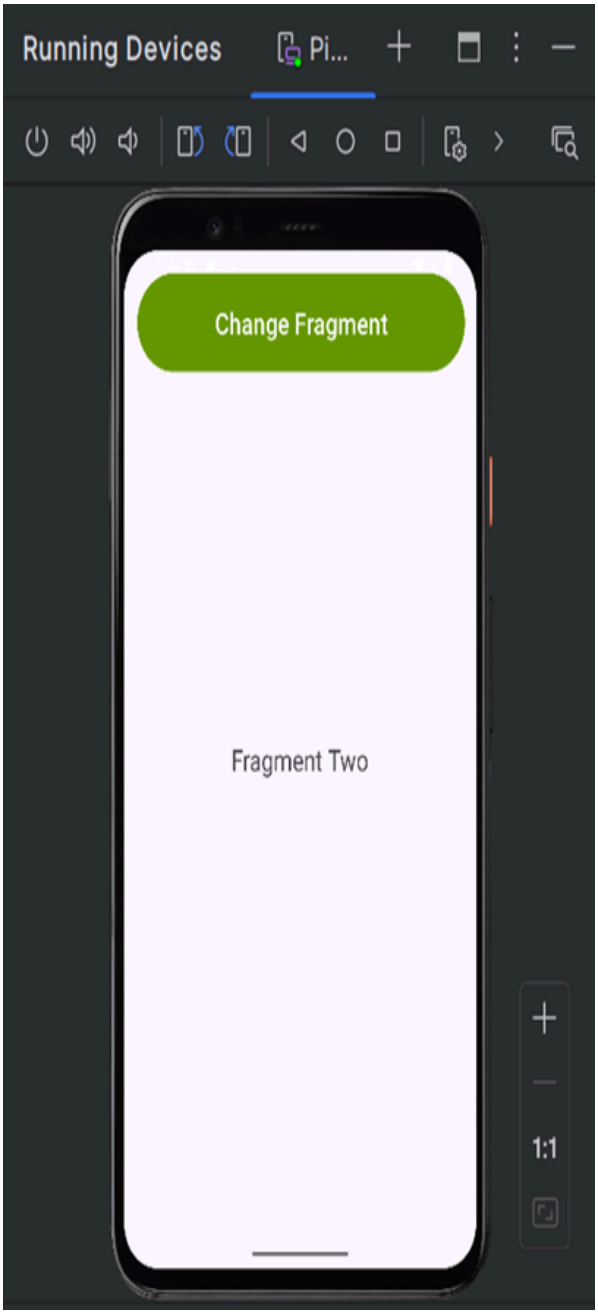
Path: app>res>layout>fragment_two.xml



Above the Android Studio near Main Menu Click the Play button to Run the Program




Running Devices
Pi...
Change Fragment
Fragment One
1:1


Running Devices
Pi...
Change Fragment
Fragment Two
1:1

6. Develop an Android application demonstrating communication between Activity and Fragment.

Project Name: ActivityFragmentCommunication

Creating Fragment
java package → New→ Fragment → Fragment (Blank)
Fragment Name: DemoFragment
Layout Name: fragment_demo

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:id="@+id/fragment_container"

   android:layout_width="match_parent"

   android:layout_height="match_parent"/>
```

Open From res/layout:
Filename: fragment_demo.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:gravity="center"

   android:orientation="vertical"

   android:padding="20dp">



   <TextView
```

```
       android:id="@+id/txtMessage"

       android:layout_width="wrap_content" android:layout_height="wrap_content"
android:textSize="22sp"

       android:text="Message from Activity"/>



   <Button

       android:id="@+id/btnSend"

       android:layout_width="wrap_content"

       android:layout_height="wrap_content"

       android:text="Send Message to Activity"

       android:layout_marginTop="20dp"/>

</LinearLayout>
```

Modify DemoFragment.java.

Filename: DemoFragment.java

```
package com.example.activityfragmentcommunication;


import android.content.Context;

import android.os.Bundle;

import android.view.LayoutInflater;

import android.view.View;

import android.view.ViewGroup; import android.widget.Button;

import android.widget.TextView;

import androidx.annotation.NonNull;

import androidx.annotation.Nullable;
```

```
import androidx.fragment.app.Fragment;

public class DemoFragment extends Fragment

{

   TextView txtMessage;

   Button btnSend;

   String activityMessage;



   FragmentCommunication communication;

public interface FragmentCommunication {

   void onMessageFromFragment(String message);

}



// Receive Activity object

@Override

public void onAttach(@NonNull Context context) {

   super.onAttach(context);

   communication = (FragmentCommunication) context;

}







// Method to receive data from Activity

public static DemoFragment newInstance(String message) {

   DemoFragment fragment = new DemoFragment();

   Bundle bundle = new Bundle();

   bundle.putString("msg", message);
```

```
    fragment.setArguments(bundle);

    return fragment;

}




@Override

public View onCreateView(LayoutInflater inflater,ViewGroup container,Bundle
savedInstanceState) {

    View view = inflater.inflate(R.layout.fragment_demo,container, false);

    txtMessage = view.findViewById(R.id.txtMessage);

    btnSend = view.findViewById(R.id.btnSend);

// Receiving data from Activity

    if(getArguments() != null){

        activityMessage = getArguments().getString("msg");

        txtMessage.setText(activityMessage);

    }



// Sending data to Activity

    btnSend.setOnClickListener(v -> {

        communication.onMessageFromFragment(

                "Hello Activity! Message from Fragment");

    }

    );



        return view;

}
```

```
}
```

Filename: MainActivity.java

```
package com.example.activityfragmentcommunication;


import android.os.Bundle;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity

       implements DemoFragment.FragmentCommunication {


   @Override

   protected void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       setContentView(R.layout.activity_main);


       // Sending data from Activity to Fragment

       DemoFragment fragment = DemoFragment.newInstance(

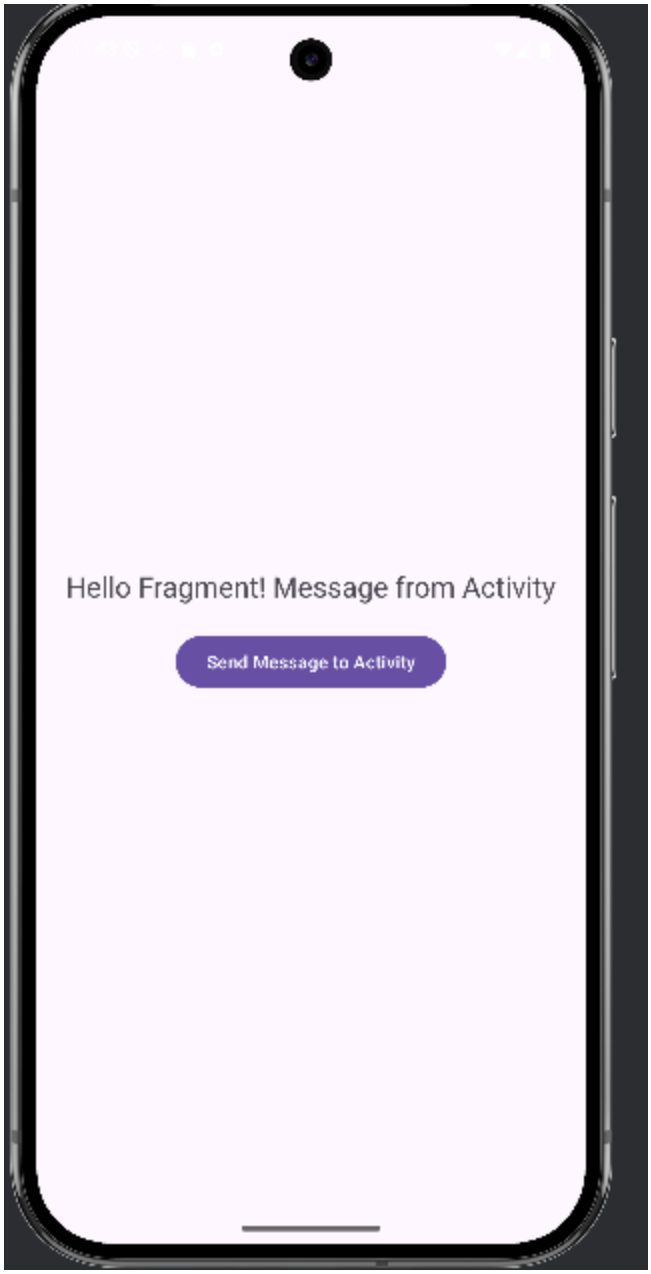
               "Hello Fragment! Message from Activity"
```

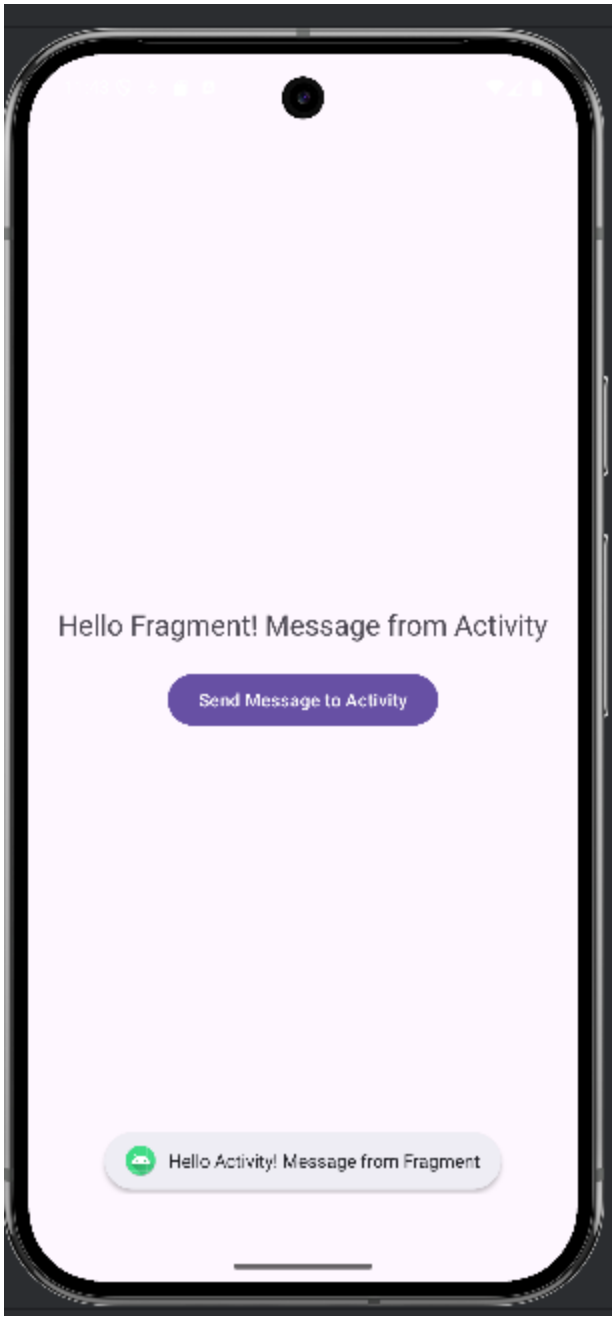
```
        );


        getSupportFragmentManager()
                .beginTransaction()
                .replace(R.id.fragment_container, fragment)
                .commit();
    }


    // Receiving data from Fragment
    @Override
    public void onMessageFromFragment(String message) {
        Toast.makeText(this, message, Toast.LENGTH_LONG).show();
    }
}
```

OUTPUT:


Hello Fragment! Message from Activity
Send Message to Activity


Hello Fragment! Message from Activity
Send Message to Activity
Hello Activity! Message from Fragment

III. Basic Layout Management

7. Develop an Android application using LinearLayout (Horizontal and Vertical) to design structured UI.

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
   xmlns:android="http://schemas.android.com/apk/res/android"
   android:id="@+id/main"
   android:layout_width="match_parent"
   android:layout_height="match_parent"
   android:orientation="vertical"
   android:padding="20dp">
   <!-- Title -->
   <TextView
       android:layout_width="wrap_content"
       android:layout_height="wrap_content"
       android:text="Student Registration" android:textSize="26sp"
       android:textStyle="bold"
       android:layout_gravity="center" android:layout_marginBottom="20dp"/>
   <!--Vertical Layout Components -->
   <EditText
       android:id="@+id/etName"
       android:layout_width="match_parent"
       android:layout_height="wrap_content"
       android:hint="Enter Name"
       android:layout_marginBottom="10dp"/>
```

```
<EditText

    android:id="@+id/etEmail"

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:hint="Enter Email"

    android:layout_marginBottom="20dp"/>


<!-- Horizontal LinearLayout -->

<LinearLayout

    android:layout_width="match_parent"

    android:layout_height="wrap_content"

    android:orientation="horizontal"

    android:gravity="center">


    <Button

        android:id="@+id/btnSubmit"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Submit"/>

    <Button

        android:id="@+id/btnReset"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Reset"

        android:layout_marginStart="20dp"/>

</LinearLayout>
```

```
</LinearLayout>
```